ELECTRODYNAMICS II · SEMESTER 5

Transients and alternating currents

Module 4 · Answer key

40 worked answers. Each answer names the prepare topic to revise first. Mark splits are suggested study rubrics.

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Section A

Q1 Model · syllabus-based

Define the LR time constant, state its unit and explain its growth interpretation.

KNOW FIRST

T1 LR current growth P1 School physics bridge
M1 Mathematics from the beginning

$\tau_L = L/R$ has units H/Ω = s. After one τ, initially zero current has risen to $1 - e^{-1} = 0.632$ of V/R. Equivalently the gap remaining to its final value is e^{-1} of its original gap.

ANSWER $\tau_L = L/R$; **63.2% rise in one time constant.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q2 Model · syllabus-based

Why does current in an ideal inductor remain continuous when a finite DC voltage is suddenly applied?

KNOW FIRST

T1 LR current growth P1 School physics bridge
M1 Mathematics from the beginning

The passive voltage is $v_L = L\,di/dt$. A finite voltage gives a finite slope, not an instantaneous finite jump. The initial current is therefore the pre-

switch value even though the applied voltage changes abruptly.

ANSWER **Finite inductor voltage implies continuous current.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q3 Model · syllabus-based

During LR decay, how much magnetic energy remains when the current is half its initial value?

KNOW FIRST

T2 LR current decay P1 School physics bridge
M1 Mathematics from the beginning

$U_L = Li^2/2$. Therefore $U/U_0 = (i/I_0)^2 = (1/2)^2 = 1/4$. The remaining three quarters has been dissipated in the resistor for the ideal closed discharge path.

ANSWER **25% remains.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q4 Model · syllabus-based

Why must the LR decay formula specify a closed discharge path?

KNOW FIRST

T2 LR current decay P1 School physics bridge
M1 Mathematics from the beginning

The formula solves $L\dot{i} + Ri = 0$ with a finite resistor path. If a switch simply opens an inductor carrying current, a large voltage can develop to maintain continuity, potentially producing arcing. That situation is not the assumed ideal R-L loop.

ANSWER **A defined resistor path is part of the exponential-decay model.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q5 Model · syllabus-based

An RC circuit has R=20 kΩ and C=50 μF. Find τ and the charging fraction at t=τ.

KNOW FIRST

T3 RC charging and discharging P1 School physics bridge M1 Mathematics from the beginning

$\tau = RC = (2.0 \times 10^{4})(50 \times 10^{-6}) = 1.0$ s. The capacitor voltage and charge reach $1 - e^{-1} = 0.632$ of their final values, while charging current falls to 0.368 of its initial value.

ANSWER **τ=1.0 s; charge fraction 63.2%.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q6 Model · syllabus-based

Which capacitor quantity remains continuous under finite current at switching, and why?

KNOW FIRST

T3 RC charging and discharging P1 School physics bridge M1 Mathematics from the beginning

Capacitor voltage remains continuous because $i = C\,dv_C/dt$. A finite instantaneous voltage jump would need an impulse of unbounded current in the ideal model. Charge q=Cv is likewise continuous.

ANSWER **Finite current implies continuous capacitor voltage and charge.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q7 Model · syllabus-based

Write the natural angular frequency and frequency of an ideal LC circuit, and distinguish the units.

KNOW FIRST

T4 Free LC oscillations P1 School physics bridge M1 Mathematics from the beginning

$\omega_0 = 1/\sqrt{LC}$ in rad/s, while $f_0 = \omega_0/(2\pi) = 1/(2\pi\sqrt{LC})$ in hertz. Confusing the two loses a factor of 2π.

ANSWER **Angular frequency and ordinary frequency differ by 2π.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q8 Model · syllabus-based

An isolated LC circuit starts with a charged capacitor and zero current. Where is its energy after a quarter-cycle?

KNOW FIRST

T4 Free LC oscillations P1 School physics bridge M1 Mathematics from the beginning

At $t = T/4$, $q = q_0 \cos(\pi/2) = 0$, so capacitor energy is zero. Current magnitude is maximal and all energy is magnetic, $Li^2/2 = q_0^2/(2C)$.

ANSWER **All stored energy is in the inductor.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q9 Model · syllabus-based

An uncharged capacitor and ideal inductor are connected in series to a DC source V at t=0. What is the maximum capacitor voltage?

KNOW FIRST

T5 LC response to a DC step P1 School physics bridge M1 Mathematics from the beginning

$v_C = V(1 - \cos \omega_0 t)$ under zero initial current and charge. At half a natural period, cosine is −1 and $v_C = 2V$. The ideal circuit does not settle because it lacks resistance.

ANSWER **Maximum capacitor voltage is 2V.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q10 Model · syllabus-based

State the peak emf of a rotating N-turn coil and explain its dependence on angular speed.

KNOW FIRST

T6 Rotating-coil AC generator P1 School physics bridge M1 Mathematics from the beginning

For area A in uniform B, $\mathcal{E}_0 = NBA\omega$. Faster rotation increases the rate of change of linked flux, so peak emf is proportional to ω. The result assumes rotation about an axis in the coil plane perpendicular to B.

ANSWER $\mathcal{E}_0 = NBA\omega$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q11 Model · syllabus-based

At what coil orientations are flux and emf maximum in an ideal AC generator?

KNOW FIRST

T6 Rotating-coil AC generator P1 School physics bridge M1 Mathematics from the beginning

Flux magnitude is maximum when the coil normal is parallel or antiparallel to B; its derivative and emf are then zero. Emf magnitude is maximum when the normal is perpendicular to B, where flux crosses zero fastest.

ANSWER **Flux and emf maxima are separated by a quarter-cycle.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q12 Model · syllabus-based

Describe voltage, current and average power for a pure resistor driven by sinusoidal voltage.

KNOW FIRST

T7 Pure R, L and C on AC P1 School physics bridge M1 Mathematics from the beginning

Current is in phase with voltage because $i = v/R$. $I_{\mathrm{rms}} = V_{\mathrm{rms}}/R$ and $P = V_{\mathrm{rms}} I_{\mathrm{rms}} = I_{\mathrm{rms}}^2 R$. Instantaneous power is nonnegative.

ANSWER **A resistor dissipates positive mean power with power factor one.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q13 Model · syllabus-based

State the reactance and phase relation for an ideal inductor on sinusoidal AC.

KNOW FIRST

T7 Pure R, L and C on AC P1 School physics bridge M1 Mathematics from the beginning

$X_L = \omega L$ and $Z_L = j\omega L$. Voltage leads current by 90°, or current lags voltage by 90°. Average power is zero because stored magnetic energy is returned each cycle.

ANSWER **Inductive current lags voltage by 90°.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q14 Model · syllabus-based

State the reactance and phase relation for an ideal capacitor on sinusoidal AC.

KNOW FIRST

T7 Pure R, L and C on AC P1 School physics bridge M1 Mathematics from the beginning

$X_C = 1/(\omega C)$ as a positive magnitude and $Z_C = -jX_C$. Current leads voltage by 90° because $i = C\,dv/dt$. Mean power is zero for an ideal capacitor.

ANSWER **Capacitive current leads voltage by 90°.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q15 Model · syllabus-based

What does multiplying a phasor by j do, and why is j used rather than i here?

KNOW FIRST

T8 Complex impedance and series RL or RC P1 School physics bridge M1 Mathematics from the beginning

Since $j^2 = -1$, multiplication by j rotates a complex-plane vector +90° without changing its magnitude. The letter j avoids conflict with the

time-dependent current i. It represents phase mathematically rather than a second physical current.

ANSWER **j denotes a +90° phase rotation.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q16 Model · syllabus-based

Why can magnitudes of resistor and inductor voltages not be added directly in a series RL circuit?

KNOW FIRST

T8 Complex impedance and series RL or RC P1 School physics bridge M1 Mathematics from the beginning

They differ in phase by 90°. With current as reference, $\widetilde{V} = V_R + jV_L$, so $V = \sqrt{V_R^2 + V_L^2}$, not $V_R + V_L$. Their instantaneous values still obey Kirchhoff's law.

ANSWER **Add voltages as phasors, then take the magnitude.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q17 Model · syllabus-based

What are ideal input current and impedance at parallel LC resonance under an AC voltage source?

KNOW FIRST

T9 Parallel LC and antiresonance P1 School physics bridge M1 Mathematics from the beginning

At $\omega_0 = 1/\sqrt{LC}$, $Y = j(\omega C - 1/\omega L) = 0$. Thus source current is zero and input impedance tends to infinity. Opposite nonzero branch currents can circulate.

ANSWER **Ideal parallel resonance gives zero input current and infinite impedance.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q18 Model · syllabus-based

State the resonance condition and input impedance of a series LCR circuit.

KNOW FIRST

T10 Series LCR and resonance P1 School physics bridge M1 Mathematics from the beginning

Resonance occurs when $\omega L = 1/(\omega C)$, giving $\omega_0 = 1/\sqrt{LC}$. Reactive impedances cancel, so Z=R, current is in phase with voltage and its amplitude is maximal at fixed V.

ANSWER **At series resonance, Z=R.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q19 Model · syllabus-based

State whether a series LCR circuit is capacitive or inductive below and above its resonance frequency.

KNOW FIRST

T10 Series LCR and resonance P1 School physics bridge M1 Mathematics from the beginning

Below resonance, $1/(\omega C) > \omega L$, so net reactance is negative and current leads. Above resonance, $\omega L > 1/(\omega C)$, so net reactance is positive and current lags.

ANSWER **Below: capacitive. Above: inductive.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q20 Model · syllabus-based

Define power factor and distinguish real power from apparent power.

KNOW FIRST

T11 Power in AC P1 School physics bridge M1 Mathematics from the beginning

Power factor is $\cos\phi$, where φ is the phase difference of voltage and current for a sinusoidal load. Apparent power is $S_{\text{app}} = V_{\text{rms}} I_{\text{rms}}$ in VA; real mean power is $P = S_{\text{app}} \cos\phi$ in W.

ANSWER **Only the in-phase contribution transfers net energy per cycle.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Section B

Q21 Model · syllabus-based

Derive current growth in a series LR circuit connected to constant V at t=0, with i(0)=0.

KNOW FIRST

T1 LR current growth P1 School physics bridge
M1 Mathematics from the beginning

1. Kirchhoff gives $L\dot{i} + Ri = V$.
   Why: Source voltage equals the two passive drops.
2. Write $\dot{i} = (R/L)(V/R - i)$.
   Why: This exposes the final value and decay coefficient.
3. The general solution is $i = V/R + Ke^{-Rt/L}$.
   Why: The deviation from the final value satisfies a decay equation.
4. At t=0, $0 = V/R + K$, so K=−V/R.
   Why: Apply the initial inductor current.
5. Check i(0)=0, i(∞)=V/R and initial slope V/L.
   Why: All three physical limits agree.

ANSWER $i(t) = (V/R)(1 - e^{-Rt/L})$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q22 Model · syllabus-based

A 12 V step drives R=6 Ω and L=3 H from zero current. Find τ, i and inductor voltage at t=0.50 s.

KNOW FIRST

T1 LR current growth P1 School physics bridge
M1 Mathematics from the beginning

1. $\tau = L/R = 0.50$ s and $I_\infty = V/R = 2$ A.
   Why: Determine time and current scales first.
2. At one τ, $i = 2(1 - e^{-1}) = 1.264$ A.
   Why: Use the growth solution.
3. $v_L = V - Ri = 12 - 6(1.264) = 4.415$ V.
   Why: The source voltage is shared between R and L.
4. Equivalently $v_L = Ve^{-1}$.
   Why: The inductor's remaining drop decays exponentially.

ANSWER **τ=0.50 s; i≈1.264 A; $v_L \approx 4.415$ V.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q23 Model · syllabus-based

Derive LR decay from initial current I_0 and show the total resistor heat equals the initial magnetic energy.

KNOW FIRST

T2 LR current decay P1 School physics bridge
M1 Mathematics from the beginning

1. $Li + Ri = 0$ gives $di/i = -(R/L)dt$.
   Why: No source remains in the closed loop.

2. Integrate and impose i(0)=I_0: $i = I_0 e^{-Rt/L}$.
   Why: The constant fixes the initial current.

3. Heat is $W = \int_0^\infty Ri^2 dt = RI_0^2 \int_0^\infty e^{-2Rt/L} dt$.
   Why: Joule power is Ri^2.

4. The integral equals L/(2R), giving $W = LI_0^2/2$.
   Why: The exponential vanishes at infinity.

ANSWER **All initial magnetic energy becomes resistor heat: $W = LI_0^2/2$.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q24 Model · syllabus-based

An LR circuit has L=2 H and R=10 Ω. How long does current take to fall to 10% of its initial value?

KNOW FIRST

T2 LR current decay P1 School physics bridge
M1 Mathematics from the beginning

1. $i/I_0 = e^{-t/\tau}$ with $\tau = L/R = 0.20$ s.
   Why: Use the current decay time, not the energy decay time.

2. Set $0.10 = e^{-t/0.20}$ and take logarithms: $\ln 0.10 = -t/0.20$.
   Why: A logarithm reverses the exponential.

3. $t = 0.20 \ln 10 = 0.4605$ s.
   Why: The negative log becomes a positive elapsed time.

ANSWER **Current reaches 10% at about 0.461 s.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q25 Model · syllabus-based

Derive q(t), i(t) and capacitor voltage when an initially uncharged series RC circuit is connected to DC V.

KNOW FIRST

T3 RC charging and discharging P1 School physics bridge M1 Mathematics from the beginning

1. $R\dot{q} + q/C = V$.
   Why: Use Kirchhoff and i=dq/dt.
2. The final charge is CV and the time constant is RC.
   Why: Compare to the standard first-order equation.
3. $q = CV(1 - e^{-t/RC})$ meets q(0)=0.
   Why: The deviation from CV decays exponentially.
4. Differentiate to get $i = (V/R)e^{-t/RC}$ and divide q by C for $v_C = V(1 - e^{-t/RC})$.
   Why: Current is charge rate and voltage is charge per capacitance.

5. Initially i=V/R while $v_C = 0$; finally i=0 while $v_C = V$.
   Why: These limits check the solution.

ANSWER **Charge and voltage grow; charging current decays.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q26 Model · syllabus-based

A 100 µF capacitor initially at 10 V discharges through 20 kΩ. Using i=dq/dt for its initially positive plate, find q, v and i after 2 s.

KNOW FIRST

T3 RC charging and discharging P1 School physics bridge M1 Mathematics from the beginning

1. $\tau = RC = 2$ s and $q_0 = CV_0 = 1.0 \times 10^{-3}$ C.
   Why: Convert microfarads to farads.

2. $q = q_0 e^{-1} = 3.679 \times 10^{-4}$ C and $v = 10e^{-1} = 3.679$ V.
   Why: The source-free charge decays.

3. $i = -q_0 e^{-1}/\tau = -1.839 \times 10^{-4}$ A.
   Why: The chosen positive plate loses charge.

4. A current arrow in the discharge direction would instead have positive magnitude 0.1839 mA.
   Why: A negative sign describes the reference convention.

ANSWER q≈0.3679 mC; v≈3.679 V; i≈−0.1839 mA with i=dq/dt.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q27 Model · syllabus-based

For charging an initially uncharged capacitor from an ideal constant source V through R, find source work, final stored energy and resistor heat.

KNOW FIRST

T3 RC charging and discharging P1 School physics bridge M1 Mathematics from the beginning

1. Total charge supplied is CV, so source work is $W_s = V \int i\,dt = V(CV) = CV^2$.
   Why: The source voltage stays constant throughout charging.

2. Final stored energy is $U_C = CV^2/2$.
   Why: Capacitor energy is the integral of its rising voltage over charge.

3. The difference $W_R = W_s - U_C = CV^2/2$ is resistor heat.
   Why: Energy is conserved.

4. The result is independent of positive R, although R changes the charging duration.
   Why: A smaller R raises current but shortens the transient.

ANSWER **Half the source energy is stored; half is dissipated in this ideal resistive charging process.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q28 Model · syllabus-based

An ideal isolated circuit has L=0.10 H, C=10 μF, initial capacitor voltage 20 V and zero current. Find q(t), i(t), frequency and peak current.

KNOW FIRST

T4 Free LC oscillations P1 School physics bridge
M1 Mathematics from the beginning

1. $q_0 = CV_0 = 2.0 \times 10^{-4}$ C and $\omega_0 = 1/\sqrt{LC} = 1000$ rad/s.
   Why: Use SI units in LC.
2. $q = q_0 \cos 1000t$ and $i = -\omega_0 q_0 \sin 1000t$.
   Why: Apply initial q and zero initial current.
3. Thus $i = -0.20 \sin 1000t$ A.
   Why: Current is dq/dt.
4. $f_0 = 1000/(2\pi) = 159.15$ Hz.
   Why: Convert angular to ordinary frequency.

ANSWER **$q = 2.0 \times 10^{-4} \cos 1000t$ C, $i = -0.20 \sin 1000t$ A; peak current 0.20 A.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q29 Model · syllabus-based

Derive the ideal series LC response to a DC step V with zero initial charge and current. Explain the voltage overshoot.

KNOW FIRST

T5 LC response to a DC step P1 School physics bridge M1 Mathematics from the beginning

1. $L\ddot{q} + q/C = V$.
   Why: The source remains connected.
2. Set x=q−CV to obtain $\ddot{x} + \omega_0^2 x = 0$.
   Why: The constant forcing shifts the centre of oscillation.
3. Since x(0)=−CV and $\dot{x}(0) = 0$, $x = -CV\cos\omega_0 t$.
   Why: Use both initial conditions.
4. Therefore $q = CV(1 - \cos\omega_0 t)$ and $i = CV\omega_0 \sin\omega_0 t$.
   Why: Differentiate charge for current.
5. At $\omega_0 t = \pi$, $v_C = 2V$ and i=0.
   Why: The ideal inductor's stored energy and source work drive the overshoot.
6. No resistor removes the oscillation energy.
   Why: The circuit does not settle in this ideal model.

ANSWER **The capacitor voltage oscillates from 0 to 2V; this is distinct from free discharge.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q30 Model · syllabus-based

A 100-turn coil of area 0.020 m² rotates at 50 revolutions/s in a 0.10 T uniform field. Find angular speed, emf peak and rms emf.

KNOW FIRST

T6 Rotating-coil AC generator P1 School physics bridge M1 Mathematics from the beginning

1. $\omega = 2\pi f = 100\pi$ rad/s.
   Why: One revolution is 2π radians.
2. $\mathcal{E}_0 = NBA\omega = 100(0.10)(0.020)(100\pi) = 62.83$ V.
   Why: Differentiate the rotating flux linkage.
3. $\mathcal{E}_{\text{rms}} = \mathcal{E}_0/\sqrt{2} = 44.43$ V.
   Why: The emf is sinusoidal.
4. The electrical frequency here is 50 Hz.
   Why: This is the ideal two-pole geometry stated in the topic.

ANSWER **ω=314.16 rad/s; peak emf 62.83 V; rms emf 44.43 V.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q31 Model · syllabus-based

Derive the impedances of an ideal inductor and capacitor using j and the common time factor $e^{j\omega t}$.

KNOW FIRST

T7 Pure R, L and C on AC P1 School physics bridge M1 Mathematics from the beginning

1. Differentiation of a sinusoidal phasor expression multiplies it by jω.
   Why: The derivative of $e^{j\omega t}$ is $j\omega e^{j\omega t}$.
2. For L, $\tilde{V} = j\omega L\tilde{I}$, giving $Z_L = j\omega L$.
   Why: Use $v = L\,di/dt$.
3. For C, $\tilde{I} = j\omega C\tilde{V}$, giving $Z_C = 1/(j\omega C) = -j/(\omega C)$.
   Why: Use $i = C\,dv/dt$ and $1/j = -j$.
4. Their angles are +90° and −90° respectively.
   Why: Multiplication by ±j rotates the phasor.

ANSWER Inductive impedance is +jωL; capacitive impedance is −j/(ωC).

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q32 Model · syllabus-based

At an angular frequency where $X_L = 40\ \Omega$, a series RL circuit has R=30 Ω and source 100 V rms. Find Z, current, phase and real power.

KNOW FIRST

T8 Complex impedance and series RL or RC P1 School physics bridge M1 Mathematics from the beginning

1. $Z = 30 + j40\ \Omega$, so $|Z| = 50\ \Omega$.
   Why: Series impedances add as complex numbers.
2. $I_{\text{rms}} = 100/50 = 2$ A.
   Why: Use the magnitude form of Ohm's law.
3. $\phi = \tan^{-1}(40/30) = 53.13^\circ$; current lags.
   Why: The impedance angle is positive.
4. $P = I^2R = 120$ W, also $VI\cos\phi = 100(2)(0.6)$.
   Why: Only R dissipates mean energy.

ANSWER **Z=30+j40 Ω; I=2 A lagging 53.13°; P=120 W.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q33 Model · syllabus-based

A series RC circuit has R=60 Ω, $X_C = 80$ Ω and source 100 V rms. Find current, phase, component voltages and power.

KNOW FIRST

T8 Complex impedance and series RL or RC P1 School physics bridge M1 Mathematics from the beginning

1. $Z = 60 - j80$ Ω and $|Z| = 100$ Ω.
   Why: Capacitive impedance is negative imaginary.
2. I=1 A rms and $\phi = -53.13°$, so current leads source voltage by 53.13°.
   Why: The source voltage phase is measured relative to current.
3. $V_R = IR = 60$ V and $V_C = IX_C = 80$ V.
   Why: They are perpendicular phasors, whose resultant is 100 V.
4. $P = I^2R = 60$ W.
   Why: The ideal capacitor has zero mean power.

ANSWER **I=1 A leading 53.13°; $V_R = 60$ V, $V_C = 80$ V; P=60 W.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q34 Model · syllabus-based

An ideal parallel LC has L=0.10 H and C=10 µF, driven at resonance by 10 V rms. Find frequency, branch currents and source current.

KNOW FIRST

T9 Parallel LC and antiresonance P1 School physics bridge M1 Mathematics from the beginning

1. $\omega_0 = 1/\sqrt{LC} = 1000$ rad/s and $f_0 = 159.15$ Hz.
   Why: Set the input susceptance to zero.
2. At resonance $X_L = \omega_0 L = 100\ \Omega$ and $X_C = 1/(\omega_0 C) = 100\ \Omega$.
   Why: The two reactance magnitudes match.
3. $\tilde{I}_L = -j0.10$ A and $\tilde{I}_C = +j0.10$ A for voltage as reference.
   Why: Inductive current lags and capacitive current leads.
4. Their sum is zero.
   Why: Kirchhoff adds branch current phasors.

ANSWER **Branch currents are 0.10 A rms in opposite phase; ideal source current is zero.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q35 Model · syllabus-based

A series circuit has R=10 Ω, L=0.10 H and C=10 μF, with 10 V rms at resonance. Find frequency, current and L/C rms voltages.

KNOW FIRST

T10 Series LCR and resonance P1 School physics bridge M1 Mathematics from the beginning

1. $\omega_0 = 1000$ rad/s and $f_0 = 159.15$ Hz.
   Why: Use the LC resonance formula.
2. At resonance Z=R=10 Ω, so I=1 A rms.
   Why: Reactive drops cancel at the input.
3. Each reactance magnitude is 100 Ω, so $V_L = V_C = 100$ V rms.
   Why: Multiply branch reactance by the common series current.
4. These two voltages are opposite phasors and sum to zero, leaving $V_R = 10$ V.
   Why: A large internal voltage does not contradict the source voltage.

ANSWER **I=1 A; each reactive element carries 100 V rms although the source is 10 V rms.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q36 Model · syllabus-based

A sinusoidal load takes 4 A rms from 230 V rms at power factor 0.75 lagging. Find real power, apparent power and phase angle.

KNOW FIRST

T11 Power in AC P1 School physics bridge M1 Mathematics from the beginning

1. Apparent power is $S_{\text{app}} = VI = 230(4) = 920$ VA.
   Why: This is the product of rms magnitudes.
2. Real power is $P = VI\cos\phi = 920(0.75) = 690$ W.
   Why: Only the in-phase part contributes on average.
3. $\phi = \cos^{-1}(0.75) = 41.41^\circ$ with current lagging.
   Why: The sign/lead description comes from the supplied power-factor qualifier.

ANSWER **P=690 W, apparent power=920 VA, lag angle≈41.41°.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Section C

Q37 Model · syllabus-based

Derive and compare LR growth/decay and RC charge/discharge, emphasizing initial conditions, time constants and limiting behaviour.

KNOW FIRST

T3 RC charging and discharging P1 School physics bridge M1 Mathematics from the beginning

1. LR obeys $L\dot{i} + Ri = V$; RC obeys $R\dot{q} + q/C = V$.
   Why: Use Kirchhoff with the element relations.
2. Each has form $\dot{y} = (y_\infty - y)/\tau$ and solution $y = y_\infty + (y_0 - y_\infty)e^{-t/\tau}$.
   Why: A deviation from steady state decays exponentially.
3. For LR, $\tau = L/R$ and $i_\infty = V/R$; for RC, $\tau = RC$ and $q_\infty = CV$.
   Why: Compare coefficients.
4. Removing the source into a closed discharge path sets the final state to zero, giving $i = I_0 e^{-tR/L}$ or $q = q_0 e^{-t/RC}$.
   Why: Stored energy drives a decaying response.
5. Inductor current and capacitor voltage remain continuous at switching under finite drive.
   Why: Their derivatives are limited by finite voltage or current.

6. During RC charging, current is $Ve^{-t/RC}/R$, while charge grows; during LR growth, current itself rises.
   Why: Current equals charge derivative only in the capacitor state equation.

7. Check t=0, t=∞ and units of τ in every solution.
   Why: These three checks catch sign and reciprocal errors.

ANSWER **LR time scale is L/R; RC time scale is RC; both are first-order responses with specified initial states.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q38 Model · syllabus-based

Develop free LC oscillation and compare it with an initially uncharged LC circuit subjected to a DC step. Explain energy in both cases.

KNOW FIRST

T4 Free LC oscillations P1 School physics bridge
M1 Mathematics from the beginning

1. Free LC gives $L\ddot{q} + q/C = 0$, so for q(0)=q_0 and i(0)=0, $q = q_0 \cos \omega_0 t$ and $i = -\omega_0 q_0 \sin \omega_0 t$.
   Why: The natural frequency is $1/\sqrt{LC}$.

2. Substituting into $U = q^2/(2C) + Li^2/2$ gives $U = q_0^2/(2C)$.
   Why: Sine and cosine squared sum to one and $L\omega_0^2 = 1/C$.

3. Energy alternates between C and L without loss.
   Why: No resistance or external source is present.

4. For a DC step the equation is $L\ddot{q} + q/C = V$, with zero initial q and i.
   Why: The source remains connected and changes the forcing.

5. The solution is $q = CV(1 - \cos\omega_0 t)$ and $i = CV\omega_0 \sin\omega_0 t$.
   Why: Shift q by CV and apply initial conditions.

6. Now the stored energy changes by source power Vi: $dU/dt = Vi$.
   Why: Multiply the circuit equation by i.

7. At half a cycle q=2CV, stored energy is $2CV^2$, exactly source work $Vq = 2CV^2$; on the return the ideal source can receive energy.
   Why: Overshoot is consistent with conservation.

ANSWER **Free LC has constant total stored energy; the driven LC step exchanges energy with the source and does not settle.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q39 Model · syllabus-based

Use complex impedance/admittance to explain and contrast series LCR resonance and ideal parallel LC resonance, including current and energy flow.

KNOW FIRST

T10 Series LCR and resonance P1 School physics bridge M1 Mathematics from the beginning

1. Series impedance is $Z_s = R + j(\omega L - 1/\omega C)$.
   Why: Common current permits impedance addition.
2. At $\omega_0 = 1/\sqrt{LC}$, $Z_s = R$, source current is maximum and in phase, and real power is V_{rms}^2/R.
   Why: Reactive voltage phasors cancel.
3. Individual L and C voltages may greatly exceed source voltage while cancelling each other.
   Why: Resonance constrains their sum, not individual values.
4. Parallel admittance is $Y_p = j(\omega C - 1/\omega L)$.
   Why: Common voltage permits admittance addition.
5. At the same ideal ω_0, $Y_p = 0$ and source current vanishes, while equal opposite branch currents remain.
   Why: Reactive current phasors cancel.

6. In both cases energy cycles between electric and magnetic stores; a real resistance causes loss and changes ideal sharpness.
   Why: Only dissipative elements consume mean power.

7. Specify the circuit and its losses before calling resonance “minimum” or “maximum” impedance.
   Why: Series and parallel results are opposite.

ANSWER **Series resonance minimizes impedance; ideal parallel resonance maximizes it, despite identical ideal LC frequency.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q40 Model · syllabus-based

Derive average power for sinusoidal voltage and current with phase difference, then discuss pure R, L, C and series LCR at resonance.

KNOW FIRST

T11 Power in AC P1 School physics bridge M1 Mathematics from the beginning

1. Let $v = V_0 \cos \omega t$ and $i = I_0 \cos(\omega t - \phi)$.
   Why: φ is the voltage lead over current.

2. Use $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$ to obtain $p = (V_0 I_0/2)[\cos \phi + \cos(2\omega t - \phi)]$.
   Why: Power is the instantaneous product vi.

3. Average over a full cycle: the second cosine contributes zero.
   Why: It completes an integer number of cycles.

4. Since $V_{\text{rms}} = V_0/\sqrt{2}$ and $I_{\text{rms}} = I_0/\sqrt{2}$, $P = V_{\text{rms}} I_{\text{rms}} \cos\phi$.
   Why: Convert peaks to rms values.

5. For pure R, φ=0 and P=I²R; for pure L or C, φ=±π/2 and P=0.
   Why: Reactive elements return the energy they store in the ideal model.

6. At series LCR resonance φ=0, so all input mean power is dissipated in R, although reactive voltages and stored energies are nonzero.
   Why: Cancellation of reactances gives unity power factor.

ANSWER **Average AC power equals rms voltage × rms current × power factor.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.